Homework 3: Solutions

STT 861 — Theory of Probability and Statistics I

Casella and Berger, *Statistical Inference*, Chapter 2, pp. 76–82

Homework problems: 1–12, corresponding to Exercises 2.2, 2.4, 2.6, 2.8, 2.9, 2.11, 2.17, 2.24, 2.26, 2.30, 2.32, and 2.39.

Page reference. Exercise 2.39 is on p. 82 of the supplied textbook; the assignment image lists p. 81.

Problem 1 — C&B Exercise 2.2, p. 76

Find the density of Y in each of the following cases: (a) $Y = X^2$, where X is uniform on $(0, 1)$; (b) $Y = -\log X$, where

$$f_X(x) = \frac{(n+m+1)!}{n!m!} x^n (1-x)^m, \qquad 0 < x < 1,$$

and m, n are positive integers; and (c) $Y = e^X$, where

$$f_X(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right), \qquad x > 0, \quad \sigma^2 > 0.$$

Each density is zero outside its stated support.

Solution

For a one-to-one differentiable transformation $Y = g(X)$, the density is

$$f_Y(y) = f_X\big(g^{-1}(y)\big) \left| \frac{d}{dy} g^{-1}(y) \right|$$

on the transformed support.

(a) $Y = X^2$. On $(0, 1)$, the inverse is $x = \sqrt{y}$ and $dx/dy = 1/(2\sqrt{y})$. Hence

$$\boxed{f_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}}$$

(b) $Y = -\log X$. The transformed support is $(0, \infty)$, the inverse is $x = e^{-y}$, and $|dx/dy| = e^{-y}$. Thus

$$\boxed{f_Y(y) = \begin{cases} \dfrac{(n+m+1)!}{n!m!} e^{-(n+1)y} (1 - e^{-y})^m, & y > 0, \\ 0, & \text{otherwise.} \end{cases}}$$

The absolute value of the Jacobian accounts for the decreasing transformation.

(c) $Y = e^X$. Since $X > 0$, we have $Y > 1$. The inverse is $x = \log y$ and $dx/dy = 1/y$, giving

$$\boxed{f_Y(y) = \begin{cases} \dfrac{\log y}{\sigma^2 y} \exp\left(-\dfrac{(\log y)^2}{2\sigma^2}\right), & y > 1, \\ 0, & \text{otherwise.} \end{cases}}$$

Values assigned to a density at individual support endpoints do not change the distribution.

Problem 2 **C&B Exercise 2.4, p. 76**

Let $\lambda > 0$ and define

$$f(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \qquad -\infty < x < \infty.$$

(a) Verify that f is a density. (b) Find $\mathbb{P}(X < t)$ for every real t when X has this density. (c) Find $\mathbb{P}(|X| < t)$ for every real t. Evaluate all integrals.

Solution

(a) Normalization. The function is nonnegative and

$$\int_{-\infty}^{\infty} f(x)\,dx = \frac{\lambda}{2}\int_{-\infty}^{0} e^{\lambda x}\,dx + \frac{\lambda}{2}\int_{0}^{\infty} e^{-\lambda x}\,dx = \frac{1}{2} + \frac{1}{2} = \boxed{1}.$$

Therefore f is a probability density.

(b) Left-tail probability. For $t < 0$,

$$\mathbb{P}(X < t) = \frac{\lambda}{2}\int_{-\infty}^{t} e^{\lambda x}\,dx = \frac{1}{2}e^{\lambda t}.$$

For $t \geq 0$,

$$\mathbb{P}(X < t) = \frac{1}{2} + \frac{\lambda}{2}\int_{0}^{t} e^{-\lambda x}\,dx = 1 - \frac{1}{2}e^{-\lambda t}.$$

Consequently,

$$\boxed{\mathbb{P}(X < t) = \begin{cases} \dfrac{1}{2}e^{\lambda t}, & t < 0, \\ 1 - \dfrac{1}{2}e^{-\lambda t}, & t \geq 0. \end{cases}}$$

Because X has a density, $\mathbb{P}(X = t) = 0$, so this also equals its cdf.

(c) Probability of a symmetric interval. If $t \leq 0$, the event $|X| < t$ is empty. If $t > 0$, symmetry gives

$$\mathbb{P}(|X| < t) = \int_{-t}^{t} f(x)\,dx = \lambda\int_{0}^{t} e^{-\lambda x}\,dx = 1 - e^{-\lambda t}.$$

Thus

$$\boxed{\mathbb{P}(|X| < t) = \begin{cases} 0, & t \leq 0, \\ 1 - e^{-\lambda t}, & t > 0. \end{cases}}$$

Problem 3 C&B Exercise 2.6, p. 76

Find the density of Y and verify that it integrates to one in each case: (a) $f_X(x) = \frac{1}{2}e^{-|x|}$ on $\mathbb{R}$ and $Y = |X|^3$; (b) $f_X(x) = \frac{3}{8}(x+1)^2$ on $(-1,1)$ and $Y = 1 - X^2$; (c) X has the density in (b), and

$$Y = \begin{cases} 1 - X^2, & X \le 0, \\ 1 - X, & X > 0. \end{cases}$$

In (b) and (c), f_X is zero outside $(-1,1)$.

Solution

When several inverse branches lead to the same value y, their density contributions must be added.

(a) $Y = |X|^3$**.** For $y > 0$, the two inverse branches are $x = y^{1/3}$ and $x = -y^{1/3}$, both having absolute derivative $1/(3y^{2/3})$. Therefore

$$f_Y(y) = \frac{f_X(y^{1/3}) + f_X(-y^{1/3})}{3y^{2/3}},$$

and hence

$$\boxed{f_Y(y) = \begin{cases} \dfrac{e^{-y^{1/3}}}{3y^{2/3}}, & y > 0, \\ 0, & \text{otherwise.} \end{cases}}$$

With $u = y^{1/3}$, so $dy = 3u^2\,du$,

$$\int_0^\infty f_Y(y)\,dy = \int_0^\infty e^{-u}\,du = \boxed{1}.$$

(b) $Y = 1 - X^2$**.** The support is $(0,1)$. For $0 < y < 1$, put $r = \sqrt{1-y}$. The inverse branches $x = r$ and $x = -r$ have absolute derivative $1/(2r)$. Thus

$$\begin{aligned} f_Y(y) &= \frac{f_X(r) + f_X(-r)}{2r} \\ &= \frac{3}{16r}\left[(1+r)^2 + (1-r)^2\right] = \frac{3(2-y)}{8\sqrt{1-y}}. \end{aligned}$$

Therefore

$$\boxed{f_Y(y) = \begin{cases} \dfrac{3(2-y)}{8\sqrt{1-y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}}$$

To check normalization, substitute $r = \sqrt{1-y}$ and $dy = -2r\,dr$:

$$\int_0^1 f_Y(y)\,dy = \frac{3}{4}\int_0^1 (1 + r^2)\,dr = \frac{3}{4}\left(1 + \frac{1}{3}\right) = \boxed{1}.$$

(c) Piecewise transformation. Both branches have image $(0,1)$, apart from endpoints of probability zero. For $0 < y < 1$, the nonpositive branch has inverse $x = -\sqrt{1-y}$, with absolute derivative $1/(2\sqrt{1-y})$. Its contribution is

$$\frac{3\big(1-\sqrt{1-y}\big)^2}{16\sqrt{1-y}}.$$

The positive branch has inverse $x = 1-y$ and absolute derivative 1, so its contribution is $3(2-y)^2/8$. Adding these contributions yields

$$\boxed{f_Y(y) = \begin{cases} \dfrac{3\big(1-\sqrt{1-y}\big)^2}{16\sqrt{1-y}} + \dfrac{3(2-y)^2}{8}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}}$$

For the first contribution, let $r = \sqrt{1-y}$; then

$$\int_0^1 \frac{3\big(1-\sqrt{1-y}\big)^2}{16\sqrt{1-y}}\,dy = \frac{3}{8}\int_0^1 (1-r)^2\,dr = \frac{1}{8}.$$

The second contribution integrates to

$$\frac{3}{8}\int_0^1 (2-y)^2\,dy = \frac{3}{8}\cdot\frac{8-1}{3} = \frac{7}{8}.$$

Thus the total integral is $\boxed{1/8 + 7/8 = 1}$. The two contributions correspond to $\mathbb{P}(X \le 0) = 1/8$ and $\mathbb{P}(X > 0) = 7/8$; neither branch should be omitted.

Problem 4 — C&B Exercise 2.8, p. 76–77

Show that each function below is a cdf and determine its generalized inverse:

$$\text{(a)}\quad F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-x}, & x \ge 0; \end{cases}$$

$$\text{(b)}\quad F(x) = \begin{cases} \frac{1}{2}e^{x}, & x < 0, \\ \frac{1}{2}, & 0 \le x < 1, \\ 1 - \frac{1}{2}e^{1-x}, & x \ge 1; \end{cases} \qquad \text{(c)}\quad F(x) = \begin{cases} \frac{1}{4}e^{x}, & x < 0, \\ 1 - \frac{1}{4}e^{-x}, & x \ge 0. \end{cases}$$

Solution

A cdf is nondecreasing, right-continuous, and has limits 0 and 1 at $-\infty$ and ∞, respectively. We use the textbook's definition in equation (2.1.13):

$$F^{-1}(y) = \inf\{x \in \mathbb{R} : F(x) \ge y\}, \qquad 0 < y < 1.$$

This definition also applies when F has a flat interval or a jump.

(a) The function is constant for $x < 0$ and strictly increasing for $x > 0$, since $F'(x) = e^{-x} > 0$ there. At 0 both one-sided limits and the value are 0, so it is continuous, hence right-continuous. Also $F(-\infty) = 0$ and $F(\infty) = 1$. It is therefore a cdf. For $0 < y < 1$, solving $1 - e^{-x} = y$ gives

$$\boxed{F^{-1}(y) = -\log(1-y), \qquad 0 < y < 1.}$$

(b) Each nonconstant piece is increasing, and the middle piece is constant. The pieces match at both breakpoints:

$$F(0-) = F(0) = \frac{1}{2}, \qquad F(1-) = F(1) = \frac{1}{2}.$$

Thus F is nondecreasing and continuous. Its limits at $-\infty$ and ∞ are 0 and 1, so it is a cdf. For $0 < y < 1/2$, the left piece gives $x = \log(2y)$. At $y = 1/2$, the set $\{x : F(x) \geq 1/2\}$ is $[0, \infty)$, whose infimum is 0. For $1/2 < y < 1$, the right piece gives $x = 1 - \log\big(2(1-y)\big)$. Hence

$$\boxed{F^{-1}(y) = \begin{cases} \log(2y), & 0 < y \leq \frac{1}{2}, \\ 1 - \log\big(2(1-y)\big), & \frac{1}{2} < y < 1. \end{cases}}$$

In particular, $F^{-1}(1/2) = 0$ despite the entire interval $[0, 1]$ having cdf value $1/2$.

(c) Each piece is increasing. At the breakpoint,

$$F(0-) = \frac{1}{4}, \qquad F(0) = \frac{3}{4},$$

so the jump is upward, preserving monotonicity. Moreover, $\lim_{x \downarrow 0} F(x) = 3/4 = F(0)$, and the function is continuous away from 0. Thus it is right-continuous, with limits 0 and 1 at the two infinities, and is a cdf. The jump corresponds to

$$\mathbb{P}(X = 0) = F(0) - F(0-) = \frac{1}{2}.$$

For $0 < y < 1/4$, solving the left piece gives $x = \log(4y)$. For $1/4 \leq y \leq 3/4$, no negative x has $F(x) \geq y$, whereas $F(0) = 3/4 \geq y$; hence the infimum is 0. For $3/4 < y < 1$, solving the right piece gives $x = -\log\big(4(1-y)\big)$. Consequently,

$$\boxed{F^{-1}(y) = \begin{cases} \log(4y), & 0 < y < \frac{1}{4}, \\ 0, & \frac{1}{4} \leq y \leq \frac{3}{4}, \\ -\log\big(4(1-y)\big), & \frac{3}{4} < y < 1. \end{cases}}$$

The generalized inverse handles the jump without requiring $F(x) = y$ to have a solution. Under the textbook's extended endpoint convention, all three examples have $F^{-1}(0) = -\infty$ and $F^{-1}(1) = \infty$.

Problem 5 C&B Exercise 2.9, p. 77

For a random variable with density

$$f_X(x) = \begin{cases} (x-1)/2, & 1 < x < 3, \\ 0, & \text{otherwise}, \end{cases}$$

find a monotone transformation $Y = u(X)$ having the uniform distribution on $(0, 1)$.

Solution

Integrating the density gives the cdf

$$F_X(x) = \begin{cases} 0, & x \le 1, \\ (x-1)^2/4, & 1 < x < 3, \\ 1, & x \ge 3. \end{cases}$$

Take $u(x) = F_X(x)$, which is nondecreasing on $\mathbb{R}$ and strictly increasing on $(1, 3)$. For $0 < y < 1$, the inverse on the support is $x = 1 + 2\sqrt{y}$, so

$$\mathbb{P}(Y \le y) = \mathbb{P}\left(\frac{(X-1)^2}{4} \le y\right) = \mathbb{P}(X \le 1 + 2\sqrt{y}) = F_X(1 + 2\sqrt{y}) = y.$$

Also, $0 < Y < 1$ almost surely. Thus

$$\boxed{u(x) = F_X(x), \qquad Y = \frac{(X-1)^2}{4} \sim \text{Uniform}(0, 1).}$$

The piecewise definition above makes u monotone on the whole real line; the quadratic expression is used only where X takes its values.

Problem 6 **C&B Exercise 2.11, p. 77**

Let X have the standard normal density $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$. **(a)** Calculate $\mathbb{E}X^2$ directly and also by integrating against the density of X^2 from Example 2.1.7. **(b)** Find the density, mean, and variance of $|X|$.

Solution

(a) Direct calculation. Since $\phi'(x) = -x\phi(x)$, integration by parts yields

$$\begin{aligned}\mathbb{E}X^2 &= \int_{-\infty}^{\infty} x^2\phi(x)\,dx = -\int_{-\infty}^{\infty} x\phi'(x)\,dx \\ &= -\big[x\phi(x)\big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty}\phi(x)\,dx = \boxed{1}.\end{aligned}$$

The boundary term vanishes because $xe^{-x^2/2} \to 0$ at both ends.

Using the density of $Y = X^2$. For $y > 0$, there are two inverse branches, $x = \sqrt{y}$ and $x = -\sqrt{y}$. Their density contributions add:

$$f_Y(y) = \frac{\phi(\sqrt{y}) + \phi(-\sqrt{y})}{2\sqrt{y}} = \frac{e^{-y/2}}{\sqrt{2\pi y}}, \qquad y > 0,$$

and $f_Y(y) = 0$ for $y \le 0$. Consequently

$$\mathbb{E}Y = \frac{1}{\sqrt{2\pi}}\int_0^{\infty} y^{1/2}e^{-y/2}\,dy.$$

Integrating by parts on $(0,\infty)$ gives

$$\int_0^{\infty} y^{1/2}e^{-y/2}\,dy = \big[-2y^{1/2}e^{-y/2}\big]_0^{\infty} + \int_0^{\infty} y^{-1/2}e^{-y/2}\,dy = \sqrt{2\pi}.$$

For the last equality, the boundary term is zero and the remaining integral is $\sqrt{2\pi}\int_0^{\infty} f_Y(y)\,dy = \sqrt{2\pi}$. Hence $\boxed{\mathbb{E}Y = \mathbb{E}X^2 = 1}$, agreeing with the direct calculation.

(b) Density and moments of $Y = |X|$. For $y \ge 0$,

$$F_Y(y) = \mathbb{P}(-y \le X \le y) = \Phi(y) - \Phi(-y) = 2\Phi(y) - 1,$$

where Φ is the standard normal cdf; $F_Y(y) = 0$ for $y < 0$. Differentiating gives

$$\boxed{f_Y(y) = \begin{cases}\sqrt{2/\pi}\,e^{-y^2/2}, & y > 0,\\ 0, & y \le 0.\end{cases}}$$

This is the half-normal density. Its mean is

$$\mathbb{E}Y = \sqrt{\frac{2}{\pi}}\int_0^{\infty} ye^{-y^2/2}\,dy = \sqrt{\frac{2}{\pi}}\big[-e^{-y^2/2}\big]_0^{\infty} = \boxed{\sqrt{\frac{2}{\pi}}}.$$

Because $Y^2 = X^2$, part (a) gives $\mathbb{E}Y^2 = 1$. Therefore

$$\boxed{\operatorname{Var}(Y) = \mathbb{E}Y^2 - (\mathbb{E}Y)^2 = 1 - \frac{2}{\pi}.}$$

Problem 7 C&B Exercise 2.17, p. 78

Find a median for each of the following densities:

$$\text{(a)} \quad f(x) = 3x^2, \quad 0 < x < 1; \qquad \text{(b)} \quad f(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}.$$

The density in part (a) is zero outside its stated support.

Solution

For a continuous distribution, a median m satisfies $F_X(m) = 1/2$: both $\mathbb{P}(X \le m)$ and $\mathbb{P}(X \ge m)$ then equal $1/2$.

(a) On $(0,1)$,

$$F_X(x) = \int_0^x 3t^2\,dt = x^3.$$

Thus $m^3 = 1/2$, and the median is

$$\boxed{m = 2^{-1/3}}.$$

(b) The cdf of this standard Cauchy distribution is

$$F_X(x) = \int_{-\infty}^{x} \frac{dt}{\pi(1+t^2)} = \frac{1}{2} + \frac{1}{\pi}\arctan x.$$

Setting $F_X(m) = 1/2$ yields $\arctan m = 0$, so

$$\boxed{m = 0}.$$

In both parts the cdf is strictly increasing on its support, so the median is unique.

Problem 8 C&B Exercise 2.24, p. 79

Calculate the mean and variance for each distribution:

$$\begin{aligned} &\text{(a)} \quad f_X(x) = ax^{a-1}, && 0 < x < 1, \quad a > 0; \\ &\text{(b)} \quad \mathbb{P}(X = k) = 1/n, && k = 1, \dots, n, \quad n \ge 1; \\ &\text{(c)} \quad f_X(x) = \tfrac{3}{2}(x-1)^2, && 0 < x < 2. \end{aligned}$$

Both continuous densities are zero outside their stated supports, and n in part (b) is an integer.

Solution

(a) For $r = 1, 2$,

$$\mathbb{E}X^r = a\int_0^1 x^{a+r-1}\,dx = \frac{a}{a+r}.$$

Therefore

$$\boxed{\mathbb{E}X = \frac{a}{a+1}, \qquad \operatorname{Var}(X) = \frac{a}{a+2} - \frac{a^2}{(a+1)^2} = \frac{a}{(a+1)^2(a+2)}.}$$

(b) The identities $\sum_{k=1}^{n} k = n(n+1)/2$ and $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$ give

$$\mathbb{E}X = \frac{1}{n}\sum_{k=1}^{n} k = \frac{n+1}{2}, \qquad \mathbb{E}X^2 = \frac{1}{n}\sum_{k=1}^{n} k^2 = \frac{(n+1)(2n+1)}{6}.$$

Hence

$$\boxed{\mathbb{E}X = \frac{n+1}{2}, \qquad \mathrm{Var}(X) = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2-1}{12}.}$$

These expressions also cover $n = 1$, when $X = 1$ almost surely and the variance is zero.

(c) Set $Z = X - 1$. Its density is $f_Z(z) = \frac{3}{2}z^2$ for $-1 < z < 1$, and zero otherwise. Oddness of z^3 gives

$$\mathbb{E}Z = \frac{3}{2}\int_{-1}^{1} z^3\,dz = 0, \qquad \mathbb{E}Z^2 = \frac{3}{2}\int_{-1}^{1} z^4\,dz = \frac{3}{5}.$$

Thus $\mathbb{E}X = 1$, $\mathbb{E}X^2 = \mathbb{E}(Z+1)^2 = 8/5$, and

$$\boxed{\mathbb{E}X = 1, \qquad \mathrm{Var}(X) = \mathrm{Var}(Z) = \frac{3}{5}.}$$

Problem 9 — C&B Exercise 2.26, p. 79

A density is symmetric about a when $f(a+\epsilon) = f(a-\epsilon)$ for every $\epsilon > 0$. **(a)** Give three examples. **(b)** Prove that a is a median. **(c)** Prove that the mean equals a whenever the expectation exists. **(d)** Show that the density $f(x) = e^{-x}$ for $x \geq 0$, and zero otherwise, is not symmetric. **(e)** Compare its median and mean.

Solution

(a) Three examples. The following densities are symmetric about a:

$$\text{Normal:} \quad f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-a)^2}{2\sigma^2}\right], \quad x \in \mathbb{R}, \quad \sigma > 0;$$

$$\text{Uniform:} \quad f(x) = \begin{cases} 1/(2b), & |x-a| < b, \\ 0, & \text{otherwise}, \end{cases} \quad b > 0;$$

$$\text{Cauchy:} \quad f(x) = \frac{b}{\pi\{b^2 + (x-a)^2\}}, \qquad x \in \mathbb{R}, \quad b > 0.$$

Each depends on $x - a$ only through its square or absolute value, so replacing $a + \epsilon$ by $a - \epsilon$ leaves its value unchanged.

(b) Median. Substitution and symmetry give

$$\int_{-\infty}^{a} f(x)\,dx = \int_{0}^{\infty} f(a-t)\,dt = \int_{0}^{\infty} f(a+t)\,dt = \int_{a}^{\infty} f(x)\,dx.$$

These two integrals sum to 1. Because a random variable with a pdf has no point mass at a,

$$\boxed{\mathbb{P}(X \leq a) = \mathbb{P}(X \geq a) = \frac{1}{2}.}$$

Hence a is a median. Symmetry alone need not make the median unique: if the density is zero on an interval around a, every point of that interval can be a median.

(c) Mean. Suppose the expectation exists as a finite value, so $\mathbb{E}|X| < \infty$. Then $\mathbb{E}|X-a| < \infty$, which justifies splitting and rearranging the integral below:

$$\begin{aligned}\mathbb{E}(X-a) &= \int_{-\infty}^{\infty} (x-a)f(x)\,dx \\ &= \int_0^{\infty} tf(a+t)\,dt - \int_0^{\infty} tf(a-t)\,dt = 0.\end{aligned}$$

Therefore $\boxed{\mathbb{E}X = a}$. The existence assumption is necessary; for example, the symmetric Cauchy density in part (a) has no finite mean.

(d) Unit exponential density. For any proposed center $a \in \mathbb{R}$, choose $\epsilon > |a|$. Then $a+\epsilon > 0$ and $a-\epsilon < 0$, so

$$f(a+\epsilon) = e^{-(a+\epsilon)} > 0 = f(a-\epsilon).$$

Thus this density is symmetric about no real number.

(e) Median versus mean. For $x \geq 0$, the cdf is $F_X(x) = 1-e^{-x}$. The median solves $1-e^{-m} = 1/2$, giving

$$m = \log 2.$$

Integration by parts gives

$$\mathbb{E}X = \int_0^{\infty} xe^{-x}\,dx = \left[-xe^{-x}\right]_0^{\infty} + \int_0^{\infty} e^{-x}\,dx = 1.$$

Since $\log 2 = \int_1^2 t^{-1}\,dt < \int_1^2 1\,dt = 1$,

$$\boxed{\text{median}(X) = \log 2 < 1 = \mathbb{E}X.}$$

Problem 10 C&B Exercise 2.30, p. 80

Find the moment generating functions corresponding to the following densities or mass function.

Solution

By definition, $M_X(t) = E(e^{tX})$, wherever this expectation is finite. In parts (a) and (b), assume $c > 0$.

(a) $f(x) = 1/c$**,** $0 < x < c$**.** For $t \neq 0$,

$$M_X(t) = \frac{1}{c}\int_0^c e^{tx}\,dx = \frac{e^{ct}-1}{ct}.$$

Since X is bounded, this expectation is finite for every real t. At $t = 0$, the defining integral equals 1, and the displayed quotient has the same limit. Thus

$$\boxed{M_X(t) = \begin{cases} \dfrac{e^{ct}-1}{ct}, & t \neq 0, \\ 1, & t = 0, \end{cases} \qquad t \in \mathbb{R}.}$$

(b) $f(x) = 2x/c^2$**,** $0 < x < c$**.** Integration by parts, for $t \neq 0$, gives

$$\begin{aligned}M_X(t) &= \frac{2}{c^2}\int_0^c xe^{tx}\,dx \\ &= \frac{2}{c^2}\left[\frac{xe^{tx}}{t} - \frac{e^{tx}}{t^2}\right]_0^c = \frac{2\{(ct-1)e^{ct}+1\}}{c^2t^2}.\end{aligned}$$

Again X is bounded, so the mgf is finite for all real t, and its value at 0 is 1. The apparent singularity at 0 is removable: $(ct-1)e^{ct}+1 = (ct)^2/2 + O(t^3)$. Therefore

$$\boxed{M_X(t) = \begin{cases} \dfrac{2\{(ct-1)e^{ct}+1\}}{c^2t^2}, & t \neq 0, \\ 1, & t = 0, \end{cases} \qquad t \in \mathbb{R}.}$$

(c) $f(x) = \dfrac{1}{2\beta} e^{-|x-\alpha|/\beta}$**,** $x \in \mathbb{R}$**,** $\alpha \in \mathbb{R}$**,** $\beta > 0$**.** Put $u = x - \alpha$ and split the integral at $u = 0$:

$$\begin{aligned} M_X(t) &= \frac{e^{\alpha t}}{2\beta} \left\{ \int_{-\infty}^{0} e^{(t+1/\beta)u}\, du + \int_{0}^{\infty} e^{(t-1/\beta)u}\, du \right\} \\ &= \frac{e^{\alpha t}}{2\beta} \left\{ \frac{1}{t+1/\beta} + \frac{1}{1/\beta - t} \right\} = \boxed{\frac{e^{\alpha t}}{1-\beta^2 t^2}}, \qquad |t| < 1/\beta. \end{aligned}$$

Both improper integrals converge precisely on this interval. At either endpoint or outside it, at least one integral diverges, so $M_X(t) = +\infty$. The formula gives $M_X(0) = 1$.

(d) $\mathbb{P}(X = x) = \binom{r+x-1}{x} p^r (1-p)^x$**,** $x = 0, 1, \ldots$**,** $0 < p < 1$**,** r **a positive integer.** The negative-binomial series $(1-z)^{-r} = \sum_{x=0}^{\infty} \binom{r+x-1}{x} z^x$, $|z| < 1$, with $z = (1-p)e^t$, yields

$$\begin{aligned} M_X(t) &= p^r \sum_{x=0}^{\infty} \binom{r+x-1}{x} \big((1-p)e^t\big)^x \\ &= \boxed{\left(\frac{p}{1-(1-p)e^t} \right)^r}, \qquad t < -\log(1-p). \end{aligned}$$

For $(1-p)e^t \geq 1$, the summands do not tend to zero, and hence the expectation diverges. In particular, 0 lies in the convergence domain and $M_X(0) = 1$.

Problem 11 C&B Exercise 2.32, p. 80

Let $M_X(t)$ be the moment generating function of X and define $S(t) = \log M_X(t)$. Show that $S'(0) = EX$ and $S''(0) = \mathrm{Var}(X)$.

Solution

Assume that the mgf is finite on an open interval containing 0. Choose $\delta > 0$ such that $M_X(\delta)$ and $M_X(-\delta)$ are finite. For $k = 1, 2$ and $|t| \leq \delta/2$, there is a finite constant C_k such that

$$|X|^k e^{tX} \leq C_k e^{\delta |X|} \leq C_k \big(e^{\delta X} + e^{-\delta X}\big).$$

The right-hand side is integrable. Differentiation under the expectation, justified by dominated convergence, therefore gives

$$M_X'(t) = E(Xe^{tX}), \qquad M_X''(t) = E(X^2 e^{tX})$$

near 0. Since $M_X(t) > 0$, ordinary differentiation gives

$$S'(t) = \frac{M_X'(t)}{M_X(t)}, \qquad S''(t) = \frac{M_X''(t)}{M_X(t)} - \left(\frac{M_X'(t)}{M_X(t)} \right)^2.$$

Using $M_X(0) = 1$, $M_X'(0) = EX$, and $M_X''(0) = EX^2$, we obtain

$$\boxed{S'(0) = EX}, \qquad \boxed{S''(0) = EX^2 - (EX)^2 = \mathrm{Var}(X)}.$$

Problem 12 C&B Exercise 2.39, p. 82

Calculate the indicated derivatives and justify all operations.

Solution

(a) $\frac{d}{dx}\int_0^x e^{-\lambda t}\,dt$**.** For a fixed real constant λ, the integrand is continuous. The fundamental theorem of calculus gives

$$\boxed{\frac{d}{dx}\int_0^x e^{-\lambda t}\,dt = e^{-\lambda x}}, \qquad x \in \mathbb{R}.$$

This includes $\lambda = 0$, when the integral is x.

(b) $\frac{d}{dx}\int_0^\infty e^{-xt}\,dt$**.** The improper integral converges precisely for $x > 0$, and then equals $1/x$. To justify differentiating under the integral at any $x_0 > 0$, restrict x to a neighborhood with $x \geq a = x_0/2 > 0$. There,

$$\left|\frac{\partial}{\partial x}e^{-xt}\right| = te^{-xt} \leq te^{-at}, \qquad \int_0^\infty te^{-at}\,dt = \frac{1}{a^2} < \infty.$$

The mean value theorem bounds the difference quotients by this same integrable function, so dominated convergence yields

$$\boxed{\frac{d}{dx}\int_0^\infty e^{-xt}\,dt = -\int_0^\infty te^{-xt}\,dt = -\frac{1}{x^2}}, \qquad x > 0.$$

(c) $\frac{d}{dt}\int_t^1 x^{-2}\,dx$**.** For $t > 0$, the interval between t and 1 contains no singularity, and x^{-2} is continuous there. The fundamental theorem of calculus (with a variable lower endpoint) gives

$$\boxed{\frac{d}{dt}\int_t^1 \frac{1}{x^2}\,dx = -\frac{1}{t^2}}, \qquad t > 0.$$

Equivalently, $\int_t^1 x^{-2}\,dx = 1/t - 1$; the formula also holds for $t > 1$ using the usual orientation of a definite integral. For $t \leq 0$, the singularity at 0 prevents the improper integral from being finite.

(d) $\frac{d}{dt}\int_1^\infty (x-t)^{-2}\,dx$**.** For $t < 1$, the integral converges and equals $1/(1-t)$. Fix $t_0 < 1$ and choose b with $t_0 < b < 1$. On a sufficiently small neighborhood of t_0, $t \leq b$, so for $x \geq 1$,

$$\left|\frac{\partial}{\partial t}(x-t)^{-2}\right| = 2(x-t)^{-3} \leq 2(x-b)^{-3}, \qquad \int_1^\infty 2(x-b)^{-3}\,dx = \frac{1}{(1-b)^2} < \infty.$$

As in part (b), this bounds the difference quotients and justifies differentiation by dominated convergence. Hence

$$\boxed{\frac{d}{dt}\int_1^\infty \frac{1}{(x-t)^2}\,dx = \int_1^\infty \frac{2}{(x-t)^3}\,dx = \frac{1}{(1-t)^2}}, \qquad t < 1.$$

For $t \geq 1$, the integrand has a nonintegrable singularity at an endpoint or in the interval, so the original integral is not finite.